

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
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Section / Batch:	AI&ML	Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

- 1) A document editor uses brute-force pattern matching to locate keywords in large research papers.
- a) Explain how the pattern matching process works.
 - b) Analyze the time complexity of this approach.
 - c) Justify the need for more efficient algorithms.

Code of Conduct

I Haresh Kumar N L (192425009) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Answer:

1) a) How Brute-Force Pattern Matching Works

Brute-force pattern matching is the simplest method used to find a pattern (keyword) inside a text.

Steps:

1. Start comparing the pattern with the text from the first character.
2. Compare each character of the pattern with the corresponding character in the text.
3. If all characters match, the pattern is found.
4. If a mismatch occurs, shift the pattern by one position to the right.
5. Repeat the process until the end of the text is reached or the pattern is found.

Example:

- Text: RESEARCHPAPER
- Pattern: PAPER

The algorithm checks:

- Compare with RESEA → Mismatch
- Shift by one position
- Compare again
- Continue until PAPER matches exactly.

b) Time Complexity Analysis

Let:

- n = Length of the text
- m = Length of the pattern

Case	Time Complexity
Best Case	$O(n)$
Average Case	$O(n \times m)$
Worst Case	$O(n \times m)$

Explanation:

- In the worst case, the algorithm compares almost every character of the pattern at every position in the text.
- Therefore, the total number of comparisons can be $(n - m + 1) \times m$, which is approximately $O(n \times m)$.

c) Need for More Efficient Algorithms

Brute-force pattern matching becomes inefficient for large research papers because it performs many unnecessary comparisons.

Reasons to use more efficient algorithms:

- Faster searching in large documents.
- Reduces execution time.
- Minimizes repeated character comparisons.
- Improves performance in text editors and search engines.
- Suitable for searching multiple keywords in large datasets.

Examples of Efficient Algorithms:

- Knuth-Morris-Pratt (KMP): Uses a prefix table to avoid rechecking characters. Time Complexity: $O(n + m)$.
- Boyer-Moore: Skips multiple characters using heuristics, making it very fast in practice.
- Rabin-Karp: Uses hashing for efficient searching of multiple patterns.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____